

IOT BASED HOME AUTOMATION

CHAPTER-1

1.1 INTRODUCTION:

Home Automation is the use of technology to control and automate various device and systems in your home. It allows you to remotely monitor and control things like lighting, temperature, security cameras, appliances, and more. With home automation, you can create schedules, set up triggers, and control your home using voice commands or mobile apps. It provides convenience, energy efficiency, and increased security by giving you the ability to manage your home from anywhere. These functions enabling users to adjust brightness and schedule lighting operations for energy efficiency; thermostat control, where smart thermostats learn user preferences and adapt to optimize comfort and cost savings; and comprehensive security and surveillance systems, incorporating smart locks, video doorbells, security cameras, and motion sensors, which bolster home security with remote monitoring capabilities. Moreover, voice assistants like Amazon Alexa and Google Assistant are integral components, providing centralized control for various home automations tasks, from adjusting temperature settings to managing entertainment systems and even locking doors.

Home automation, often referred to as smart home technology, is the application of advanced technology, is the application of a residence, enhancing convenience, efficiency, and security in daily life. At its core, home automation involves the interconnection of devices and systems within a household through a central hub, mobile applications, or voice-activated commands.

1.2 PURPOSE OF HOME AUTOMATION

The purpose of a home automation system is to streamline how your home functions.

Consider some of these benefits,

- Remote access – Control your home from mobile devices, including your laptop, tablet, or smartphone.
- Comfort – Use home automation to make your home a more comfortable, livable space.
- Convenience – Program devices to turn on automatically at certain times, or access their settings remotely from anywhere with an Internet connection. When you don't have to remember to lock the door behind you or switch off the lights, you can turn your attention to more important things.
- Increased safety – Smart fire detectors, carbon monoxide monitors, pressure sensors, and other home automation security features can help protect your home from disasters.
- Energy efficiency – Home automation allows you to be more mindful of your power usage. For example, you can save on energy bills by reducing the length of time that lights stay on, or by lowering temperatures when you leave a room.

1.3 ADVANTAGES OF HOME AUTOMATION

- Easy to access i.e. user friendly with a lot of features.
- The controller is efficient as it has a dual-core with BLE and WiFi support.
- The standalone system has a different mode as per need.
- To control appliances you get manual and internet mode.
- Google Assistant/ Amazon Alexa/ smart things support which makes it even smarter.
- Even when your far away from your home appliances, you will able to operate the devices if and only if the ESP32 is connected to Wi-Fi.

1.4 DISADVANTAGES OF HOME AUTOMATION

- Smart assistants can sometimes misinterpret voice commands or experience downtime, leading to frustration.
- Smart devices can be vulnerable to hacking, potentially exposing your home to security risks if not properly secured.
- Without an internet connection, Google Assistant and Alexa may have limited functionality, whereas manual switches always work.

1.5 APPLICATIONS OF HOME AUTOMATION

Home automation using Google Assistant, Alexa, and manual switches has various applications:

- Google Assistant and Alexa can control audio and video devices, streaming services, and smart TVs, while manual switches can turn on/off entertainment systems.
- Google Assistant and Alexa enable hands-free control for various tasks, like setting reminders, checking the weather, or answering questions

CHAPTER-2

2.1 OVERVIEW

LITERATURE SURVEY

Most of the devices in our home are not interconnected. Due to this factor, we cannot operate them from remote locations. Whenever any family going on a long journey for a few days then keeping eye on the home is not possible. Even sometimes we forget to turn off our devices, for example, if we forget to turn off home lights, so at that time you can't turn off your lights till your no at home. Due to this you have to pay extra electricity bill. And

sometimes it leads to device failure. This may be due to overheating and the short circuit of the device connections.

Many existing, well established home automation system are based on wired communication. This does not pose a problem unit the system is planned well in advance and installed during the physical construction of the building. The existing system has the above drawbacks and to overcome this, different proposed systems has been introduced.

2.2 HISTORY

The dictionary defines automation as “the technique of making a process or a system operate automatically.” We define automation as “the creation and application technology to monitor and control the production and delivery of products and services.” Using our definition, the automation professional is “any individual involved in the creation and application of technology to monitor and control the production and delivery of products and services.”

Nikhil Rathod, proposed flexible home Automation system which is using advanced versions of the Arduino microcontrollers. They concluded that it will be very easy to use these advance versions of Arduino with easy coding. We can manage the complete home domestic devices over the internet. This augments the reassure ability of humans and it will decrease the Human hard work.

Prasad Mhanta, presented a suggestion for home automation by means of Google Assistant Home mechanization. This can be promising to build up a self-governing and self- managing system and extensible home coordination that can hold today’s elegant devices and technologies of conflicting characteristic and protocols. They suggest that residence appliances can independently be controlled equally from inside the home and distantly. This is very obliging to physically challenged citizens.

Sandeep Chintla, presented a study on IoT Technology which is flattering more resourceful the days because of marvelous increase in home mechanization applications and as well all could be controlled from wherever sitting at a place. The running of these IoT devices is mechanical and there is no want foe man-made invention. It provides better advantages which decrease power by civilizing home security. On the other hand, house automation scheme is elastic to lodge new appliances because one can function the appliances even although if the user is far away from the house. With the aid of such IoT devices, the everyday lives or the job of the user is made easier and accurate.

2.3 IOT APPLICATION ESP RAIN MAKER:

ESP Rain Maker is a light-weight A IoT Cloud software, fully integrated into the AWS serverless architecture, which allows customers to build, develop and deploy customized A IoT solutions with a minimum amount of code and maximum security. Here we should download the ESP Rain Maker app, where we can scan. After scanning, in the Home page we can able to see the home apparatus which we want to control (lights).

In this app we can rename the home apparatus according to our convenience instead of renaming in the code. The main feature of this app is that, we can have the access upto 8 switches. But here in our project, since we are using 4 channel relay module, we are using access for 4 switches. And another benefit is that, there is no need of using the same Wi-Fi credentials which we have used for ESP32 module. Here we can change the Wi-Fi credentials on your android device in which the app is installed and accordingly we can control the home appliances through voice commands.

2.4 GOOGLE ASSISTANT

The Google Assistant is a virtual assistant software application developed by Google that is primarily available on mobile and home automation devices. Based on artificial intelligence, The Google Assistant can engage in two-way conversations, unlike the company's previous virtual assistant. The Google Assistant debuted in May 2016 as part of Google's messaging app Allo, and its voice-activated speaker Google Nest. After a period of exclusivity on the Google Pixel smartphones, it was deployed on other Android devices starting in February 2017, including third-party smartphones and Android Wear (now Wear OS), and was released as a standalone app on the iOS operating system in May 2017. Alongside the announcement of a software development kit in April 2017, Assistant has been further extended to support a large variety of devices, including cars and third-party smart home appliances. The functionality of the Assistant can also be enhanced by third-party developers.

Now Assistant uses what you say, and info from your linked devices and services, to understand and respond to you. It also uses this info for other purposes stated in Google's Privacy Policy, such as improvements to Google services. Google Assistant is ready to help you get things done, anytime, anywhere. When you ask a question or tell it to do something, Assistant wants to respond to your request in the most helpful way possible - whether you want assistance with everyday tasks, controlling smart home devices, enjoying music or games, communicating with friends and family, getting quick answers or local information, or many other things.

Google Assistant enables voice-forward control of Android apps. Using Assistant, users can launch apps, perform tasks, access content, and more by using their voice to say things like, "Hey Google, start a run, on Example App." Google Assistant can interact with your Android phone to do a variety of tasks, such as setting alarms or playing music. Like Siri, it can even handle some home automation devices. Google has a page explaining various types of actions here. Like Siri, you can ask Google Assistant general questions. Assistant operates by using advanced natural language processing and artificial intelligence technologies. When activated by the wake phrases "Hey Google" or "OK Google," it processes spoken commands through speech recognition, converts them into text, and employs natural language processing to discern context and intent. With access to Google's extensive Knowledge Graph, it can fetch information from the web and interact with third-party services via APIs. The assistant generates responses, which can be spoken or displayed as text, and may take actions to fulfill tasks. Through machine learning, it continuously improves its understanding of user preferences. User privacy is prioritized, with options to review and delete data. Available on a variety of devices, Google Assistant provides a versatile and interactive experience.

ALEXA:

For Alexa, the voice of speaker is recorded and sent to Amazon's servers to analyze which words most closely correlate with the sounds spoken. Once the command is understood, Amazon then sends a requested information back to the device, and in some cases, alexa will speak a response. Here main automatic speech recognition and natural language understanding functions break down the command. This data gets shifted through algorithms to determine the inputted phrase meaning. A response is then given back to user. The base components of Alexa, from a user perspective, are an Amazon account and an Alexa – enabled device connect to the internet, usually a smart speaker or display. Alexa devices listen for voice commands, upload them to Amazon's servers for translation, then deliver results in the form of audio, video, or device or accessory triggers.

To convert the audio into text, Alexa will analyze characteristics of the user's speech such as frequency and pitch to give you feature values. A decoder will determine what the most likely sequence of words is, given the input features and the model, which is split into two pieces. The first of these pieces is the prior, which gives you the most likely sequence based on a huge amount of existing text, without looking at the features, the other is the acoustic model, which is trained with deep learning by looking at pairings of audio and transcripts. These are combined and dynamic coding is applied, which had to happen in real time.”

2.5 INTERFACING WITH ESP32

The ESP32 is an open-source hardware platform based on a dual-core Xtensa LX6 microcontroller developed by Express if Systems. It was first introduced in 2016 and has since gained popularity for its versatility and robust capabilities. The microcontroller architecture of the ESP32 is based on Reduced Instruction Set Computing (RISC), similar to the Atmel AVR microcontroller used in Arduino. RISC architecture prioritizes simplified instructions for faster execution, contributing to the ESP32's high performance. Official ESP32 development boards typically include features like built-in Wi-Fi and Bluetooth connectivity, making them suitable for a wide range of IoT applications. Additionally, the ESP32 microcontroller comes pre- programmed with a bootloader, simplifying the process of uploading programs to the onboard flash memory. This feature streamlines development and enables users to leverage an ordinary computer for programming tasks.

Like the Arduino, the ESP32 features complementary components to facilitate programming and integration into various circuits. It offers a standard layout for connectors, allowing for easy attachment of add-on modules and accessories. While shields are commonly associated with Arduino, the ESP32 ecosystem includes expansion boards and modules that serve similar functions. These modules can communicate directly with the ESP32 via various pins or through protocols like I²C and SPI. In terms of I/O capabilities, the ESP32 provides numerous digital and analog pins for interfacing with external devices and sensors. These pins can be utilized for a variety of purposes, including reading sensor data, controlling actuators, and communicating with other microcontrollers or peripheral devices. Overall, the

ESP32 offers a powerful and flexible platform for building IoT projects, prototyping, and exploring various embedded applications.

ESP32 is the series of low-cost, low power system on chip microcontrollers with integrated Wi-Fi and dual-mode Bluetooth. The ESP32 is the name chip that was developed by Espressif Systems. We can program the ESP32 in variety of languages and Programming environments including C/C++, Micropython, Lua. Typically Wi-Fi range is 500 meters to 200 meters depending on the antenna. ESP32 is better than ESP8266(Node MCU) as it provides you with faster processor and good memory size. ESP32 provides you with reliable and hi-tech security. The ESP32's operating voltage range is 2.2 to 3.6V. Under normal operation the ESP32 Thing will power the chip at 3.3V. The I/O pins are not 5V-tolerant! If you interface the board with 5V (or higher) components then 7805 regulator used. It is low cost and highly versatile microcontroller used for various applications which include wireless communication, IoT (Internet of things) devices, home automation, robotics, embedded systems.

CHAPTER-3

HOME AUTOMATION SYSTEM

3.1 METHODOLOGY

The methodology of this project design includes implementation of the proposed method.

The basic steps involved are discussed above. Here we have divided into two main categories.

- Hardware
- Software

Implementation:

By considering this block diagram, to convert AC supply to DC supply we use a 5V 1A Power Bank (Power Supply) which provides DC power to the circuit to get desired output. If there is an internet issue, we can control this system using manual switches.

3.2 BLOCK DIAGRAM

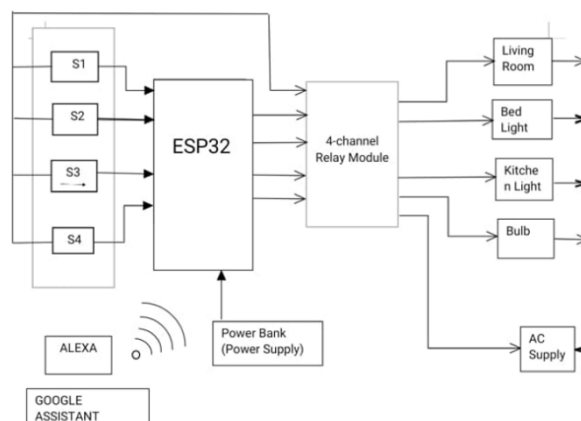


Fig. 3.1 Block diagram

3.3 COMPONENTS

- ESP 32 Dev Module
- 4 Channel Relay Module

ESP32 Dev Module :

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4-Channel Relay Module:

The 4 Channel Relay Module is a convenient board which can be used to control high voltage, high current load such as motor, solenoid valves, lamps and AC load. It is designed to interface with microcontroller such as Arduino, PIC. It allows you to control multiple electrical circuits using a single control signal. With a 4-channel relay module, you can easily switch on and off different circuits using your chosen control method, such as manual switches or voice commands.

In general, a 4-channel relay module is a device that has four separate relays built into it. Each relay is like a switch that can control the flow of electricity to a different circuit. The module usually has input pins where you can connect control signals, such as from a microcontroller or a home automation system like Google Assistant or Alexa. When the control signal is activated, the corresponding relay switches on or off, allowing or blocking the flow of electricity to the connected circuit. This way, you can control up to four different circuits independently using the module. It's a convenient and versatile component for various home automation projects.

CHAPTER -4

ANALYSIS AND OBSERVATION

4.1 DISCUSSIONS

In this present proposed system related to this project, the ESP32 Dev module be supposed to be included with normal home apparatus. The stream printed on Arduino software will be convey in the ESP rainmaker application that will be synchronized with the Amazon's Alexa and Google Assistant. At whatsoever point orders are specified it triggers the encoding interface call during the cloud to the synchronized ESP32. The information is transformed into sign to roll the home apparatus ON or OFF. The tip of the task to recommend a cost creative voice controlled (Google Assistant) home mechanization controlling universal

apparatus originate in one's home. And the best advantage in this is there will no need for the Amazon's Echo Dot for the controlling of the home apparatus, only the applications of Alexa and Google Assistant will be required, in which Google Assistant is inbuilt application in android phones.

4.2 HARDWARE IMPLEMENTATION

It has the capability to connect to the router. It would also be able to turn on/off specified devices, such as lights and fans. It is called the 'Control Unit'. In this project the hardware components are

Fig 4.1 4-Channel Relay

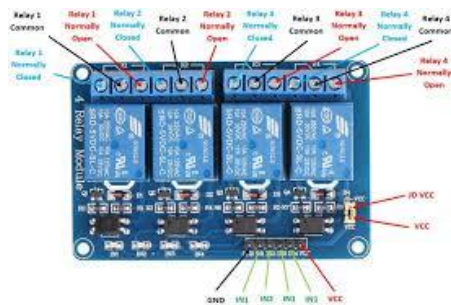


Fig 4.2 ESP32 Module



Fig 4.3 Circuit Connections



Fig 4.4 Output



4.3 SOFTWARE IMPLEMENTATION

The ESP Rainmaker app, Alexa app and Google Assistant constitute the software of design and this application would be integrated in the android devices. In Arduino IDE Software the code will be executed. The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them.

- Pin mode () {

```
// used to set the input and output mode
```

```
}
```

Example – pin mode (13, OUTPUT)

- Digital Read () {

```
// Reads the value from a specified digital pin i.e. reads the value from input pin and assigns it to value.
```

```
}
```

- Setup () – it is function used to initiate pin modules, start using libraries.

4.4 RESULT

First we observed the turning ON and OFF of the devices in the Arduino IDE software. If the device is turned ON then it shows ‘true’ and if device is turned OFF then it shows ‘false’. As the system has tons of feature we can control devices using a voice assistant like Google Assistant and Alexa applications. When we want to control any device we give a command or manually switch the devices then ESP32 gets the command from the user and send to the IoT platform in our case which is ESP Rain Maker. Google Assistant-After linking to the Google Assistant through the IoT Platform on Google Home or Google Assistant application the devices can be operated .By this way automatic home automation can be done.

CHAPTER-5

5.1 Conclusion

CONCLUSION

In this project, we have implemented a smart home automation system that can be controlled using voice commands through Google Assistant and Alexa, as well as through traditional manual switches. The system has been experimentally tested and proven to work effectively by connecting it to simple household appliances, which were successfully controlled remotely over the internet.

The designed system goes beyond simple on/off control—it not only monitors real-time data from various sensors, including temperature, gas, light, and motion sensors, but also responds intelligently by activating or deactivating appliances based on the current conditions. For example, lights can turn on automatically when motion is detected, fans can adjust based on temperature, and alerts can be triggered if gas levels rise beyond safe limits.

This integration of IoT technology with voice control and sensor-based automation ensures a convenient, energy-efficient, and safe home environment. By combining manual control, voice commands, and automatic responses, the system provides a flexible and reliable solution for modern smart homes.

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